

Key Properties

- **Nonnegativity:** $D_f(\alpha||\beta) \geq 0$.
- **Identity:** $D_f(\alpha||\beta) = 0 \iff \alpha = \beta$.
- **Asymmetry:** In general $D_f(\alpha||\beta) \neq D_f(\beta||\alpha)$.
- **No triangle inequality:** Not a metric in general (except TV and Hellinger).
- **Convexity:** Jointly convex in (α, β) .
- **Dual form** (via convex conjugate f^*): $D_f(\alpha||\beta) = \sup_g \{ \mathbb{E}_\alpha[g] - \mathbb{E}_\beta[f^*(g)] \}.$

Use Cases in ML